

- The ***powder characteristics influence the properties of the final product.***
- Powders used in P/M are characterized by several features and each characteristic has its own influence on processing as well as on the composition, microstructure and final properties of the product.

(i) *Material characteristics* like physical and chemical characteristics. Physical characteristics include structure, porosity and density. Chemical characteristics include purity and composition.

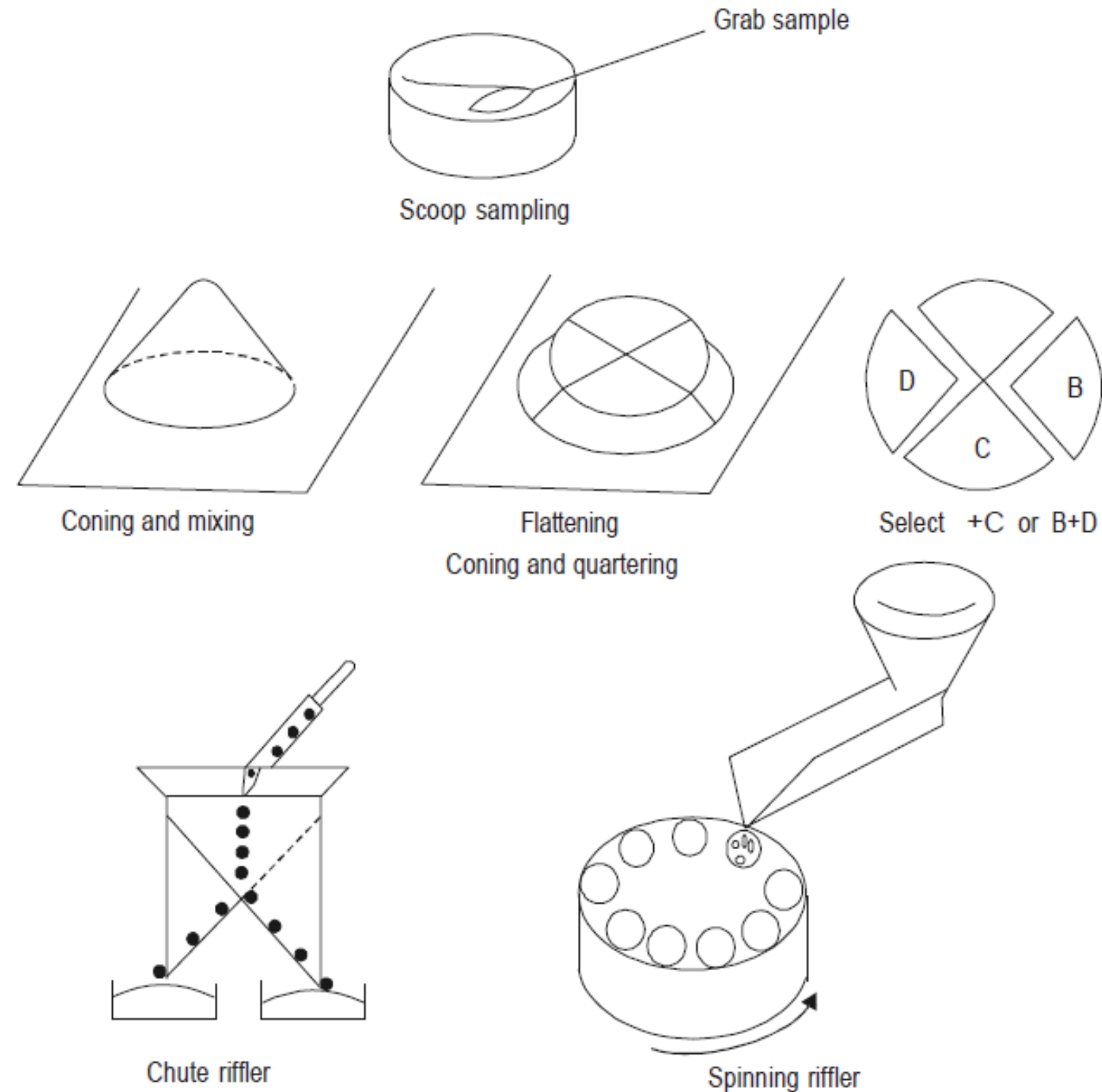
(ii) *Morphological characteristics* like shape, size, texture and surface area

(iii) *Bulk or behavioural characteristics* like flow rate, apparent density, bulk density, compressibility and green strength.

SAMPLING

- Sampling is the first activity in the characterization of metal powders.
- It is essential to select a representative sample before a metal powder lot is analysed.
- For most chemical analysis, a small quantity of sample is sufficient.
- Other measurements like sieve analysis usually need a considerable amount of powder sample for characterization.
- In any case, it is important to ensure that such a sample is fully representative of the entire range of particle sizes in the lot.
- The properties of metal powders being **statistical in nature**, their measurement requires proper sampling technique/procedures.
- **Two important rules** for obtaining good powder samples are:
 - (i) To do sampling while in motion, and
 - (ii) To obtain a large number of small lots of sample over short intervals of time.

Some important sampling techniques



- **Scoop sampling**, consists of inserting a scoop into a thoroughly mixed powder in a container and withdrawing a scoopful of powder as a sample.
- The second technique is known as **coning and quartering**.
- In this technique, powders are poured on to a polished sheet of aluminium in the form of a cone.
- The cone is then flattened and split into four equal segments, with thin sheets of polished brass and a suitable quantity for testing is collected.
- The third procedure, called **chute riffler method**, consists of a V shaped trough with a series of chutes at the bottom, which alternatively directs powders to two containers placed on either side of the trough allowing two identical samples to be collected simultaneously.
- The fourth technique, a variation of previous one, called **spinning riffler method**,
- Consists of a closed ring of containers spinning under a steady stream of powder feed so that each container collects a series of small portions of the powder feed as it passes several times beneath the powder feed.

- **Stream sampling** is another procedure commonly adopted for selecting a representative powder sample from powders.
In this technique, a container is passed through a stream of flowing powder a number of times and a representative sample is taken from the collected powder.
In the case of **bulk powders, finer particles segregate towards the bottom of the Container during transportation and handling.**
Hence a true sample must include powders from all depths.
One method of achieving this is by **inserting a sampler consisting of an inner and outer tube with a closed end and longitudinal windows.**

- This sampler, **called a thief**, is inserted into the container vertically and the sample is taken at a location about three quarters along the radius from the center of the container.
- The **inner tube of the thief is rotated**, to open the windows until the tube is filled with powder then rotated, so as to close the window.
- The thief is removed and its contents emptied through the open top into the blender, so as to mix the powders thoroughly before selecting a representative sample for analysis.

METAL POWDER CHARACTERIZATION

- (i) chemical characteristics such as composition and purity.
 - (ii) physical characteristics like size, shape and density, and
 - (iii) processing properties like flow rate, bulk density and compressibility.
-
- Standard test procedures have been prepared by Metal Powder Industries Federation (**MPIF**), **USA** and the American Society for Testing of Materials (**ASTM**).

Chemical Composition Analysis

- Chemical analysis, in addition to the elements present, must also give the amount of **metallic and non-metallic impurities**.
- The impurities can be in elemental form, or in dissolved form as solid solution or as compounds.
- The purity of a powder, the type and amount of impurities indicated by compositional analysis depends strongly on the production method.
- For instance, **electrolytic powders** may contain **dissolved gases** (oxygen, hydrogen) as impurity.
- In the case of **atomized powders**, eroded **oxide particles** from refractory nozzles/crucibles can also be present.
- In addition, the powders may absorb **moisture during storage as well as handling and form surface oxides**.

- These must also be taken into account while analyzing for composition.
- **Purity of powders containing oxides can be improved by annealing in a reduced atmosphere like hydrogen** at temperatures between 800°C and 1,000°C.
- The techniques to analyse the **different forms of impurities are different.**
- For example, the technique used to detect silicon may not be suitable or accurate if the silicon is present in the form of a compound (silica) rather than in solid solution.
- If the element is present in solid solution, then it may sometimes undergo chemical reactions during sintering.

Hydrogen loss method

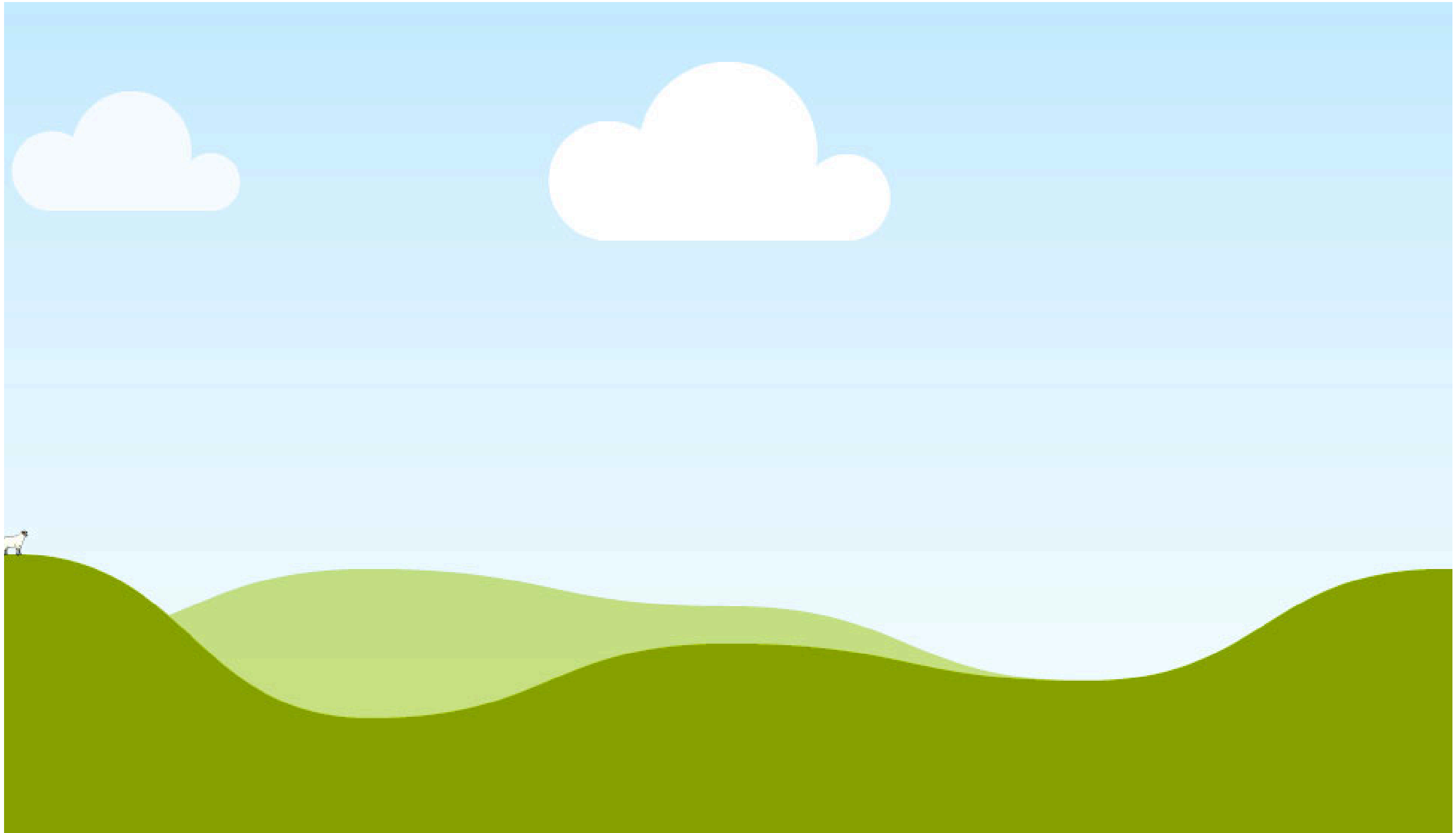
- Oxygen content of powders is measured by hydrogen loss method.
 - To measure the hydrogen loss, a 5 g-sample powder is placed in a combustion boat and exposed to a specific period of time and temperature in an atmosphere of dry, purified hydrogen.
 - The weight loss of the powder specimen due to the hydrogen exposure is measured

Standard Conditions for Hydrogen Loss Determination for Various Metals



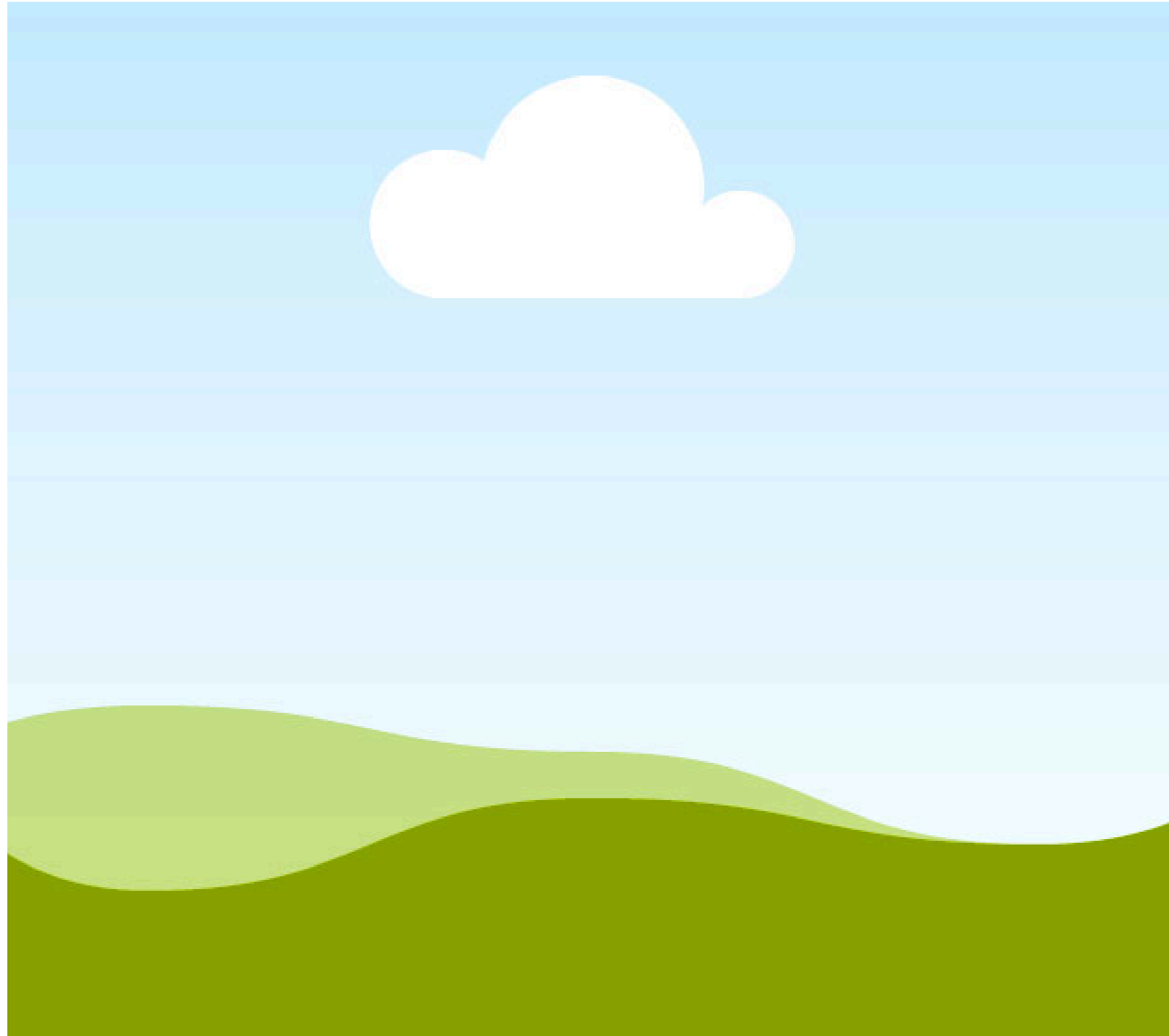
- Oxygen content of powders like iron, copper, nickel, tungsten and molybdenum can be estimated by the hydrogen loss method.
- **Stable oxides like alumina and chromia will not be reduced by hydrogen.**
- **Impurities like carbon and sulphur are estimated using carbon and sulphur analyzer.**
- Although many analytical techniques are available for determining the chemical composition of powders, the most frequently used methods are given in Table.
- In many cases, more than one method may be necessary to characterize the powders fully.

Various Chemical Characterization Methods



Surface Chemical Analysis of Metal Powders

- Elemental analysis of the first few atomic layers of a powder sample.



- The application of these techniques has resulted in the understanding that a powder particle, **though homogeneous within its bulk can be inhomogeneous near its surface.**
- Such a surface inhomogeneity is due to the different powder processing methods.
- Hence the application of surface analysis techniques to P/M products enables an improved understanding of both the surface chemistry of the **powder particles as well as the consolidated powder products and** its effect on the resulting engineering properties.

Particle Shape Analysis

- Particle shape can be defined as, “the recognized pattern of relationship between points that form the external surface of the particle.”
- Particle shape analysis is a fundamental characteristic of powder particles, which influences other parameters like flow rate, apparent density, compressibility and sinterability.
- Two different concepts used to describe and differentiate particle shapes are based on:
 - i. the dimensionality of the particle, and**
 - (ii) the surface contour of the particle.**

Four different classes of Shape characterization

- i. Verbal expression (spherical, elongated, irregular).
- ii. Use of dimensionless number (shape factor).
- iii. Shape generation using Fourier analysis, polynomial fitting, matrix mapping.
- (iv) Evaluation of bulk properties.

Verbal expression

A. One-dimensional (large aspect ratio)

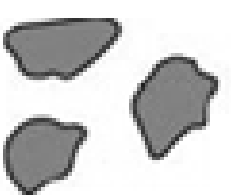
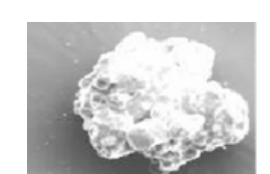
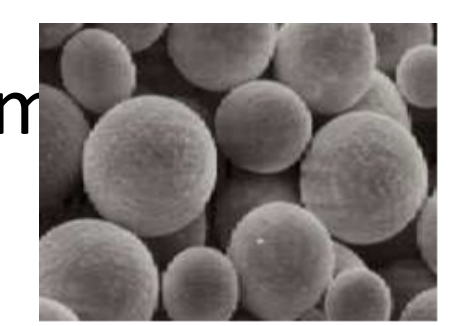
- (i) Acicular — Chemical decomposition
- (ii) Irregular, rodlike — Chemical decomposition and mechanical comminution

B. Two-dimensional (length and width much greater than the thickness)

- (i) Dendritic — Electrolytic silver, copper and iron
- (ii) Flake — Ball-milled copper, aluminium

C. Three-dimensional

- (i) Spherical — Carbonyl iron powder, condensed latex, precipitation from aqueous solution
- (ii) Angular — Comminution and carbonyl nodules
- (iii) Irregular — Atomization, chemical decomposition, reduction
- (iv) Porous — Reduction of oxides
- (v) Rounded — Atomized aluminium, copper, tin and zinc. Chemical decomposition



- The particle shape plays an important role on the final properties of the component.
- **Irregular-shaped particles**, for example, have reduced apparent density and flow rate but good pressing and sintering properties.
- **Spherical particles**, on the other hand, have maximum apparent density and flow rate but reduced pressing and sintering characteristics.
- **Dendritic powders** also have poor apparent density and poor flow rate. Among the various particle shapes, spherical shape is the ideal one.
- **Particle shape can be determined by several methods.**
- Optical microscopy, Scanning Electron Microscope (SEM), Transmission Electron Microscope (TEM) as well as Image Analyzer are all used to evaluate particle shape.

Particle Size

- Particle size can be defined as, “**the total space occupied by a powder particle.**”
- The particle size has a great importance in P/M because it affects some of the important properties such as green strength, density of compact as well as agglomeration, porosity, expulsion of trapped occluded gases, dimensional stability, flow and mixing characteristics.
- **Fine metal powders result in poor apparent density, poor flow rates and maximum sinterability while coarse powders give good apparent density, good flow rate but minimum sinterability.**

- In P/M processing, the powders used range in sizes from 0.1 to 1,000 microns in diameter.
- The powders produced by any method will have a range of sizes, ranging from about a mm to several microns.
- It is possible to maintain a narrow size distribution by close control of process parameters; however, **it is not possible to have a single particle size in a powder lot.**
- In the case of **spherical powders**, particle sizes are characterized by their **diameters**.
- Particle size and size distribution of **irregularly-shaped powders, a measurement of a certain combination of powder particle properties** such as length, volume, mass, settling rate, total or projected surface area, light scattering and electric or field perturbation is done.
- Thus, a single size measure might be misleading.
- **In general, a characteristic size for each powder is usually linked to its mass or its cross-sectional size.**

- Particle sizes can be measured either based on dimensions such as diameter or by measuring the mass, settling rate or volume.
- The definition of particle size may differ depending on the technique used, e.g. particle size may be expressed as **circle area, equivalent diameter, Stoke's diameter and intercept.**
- The **method of measuring the particle size may also differ**: whether mode, median or mean size is used in measurement, must be clearly stated.
- Also different techniques measure different distributions—**the most important ones being the number and volume distribution.**

Characteristics of Various Particle Classification Methods

Method	Size definition	Limiting value	Remarks
Sieving	Minimum feret	> 1–5 mm	Bias towards smaller size dependent on procedure
Microscopy	Nearly free choice	> 1 nm	Bias towards larger size
Electrical pulse	Cross-section equivalent	0.5–100 mm	Particles assumed to be spherical
Light scattering	Statistical	< 2 mm	Particles assumed to be spherical
Specific surface	Statistical	< 2 mm	Primary particles
Sedimentation	Stokes	0.3–100 mm	Agglomerate

- Metal powders are divided into three classes namely: (i) sieve, (ii) subsieve, (iii) submicron, or ultra-fine powders.
- The screen with the opening of finest standard mesh sieve, for production purpose is the 325-mesh screen having an aperture of 44 micron.
- **Subsieve particles are smaller than the aperture of the above screen but greater than 1 micron.**
- Majority of metal powders used in P/M industry vary in size between 4 and 200 microns.

Typical Methods of Particle Size Measurement and Their Ranges

Sieve analysis

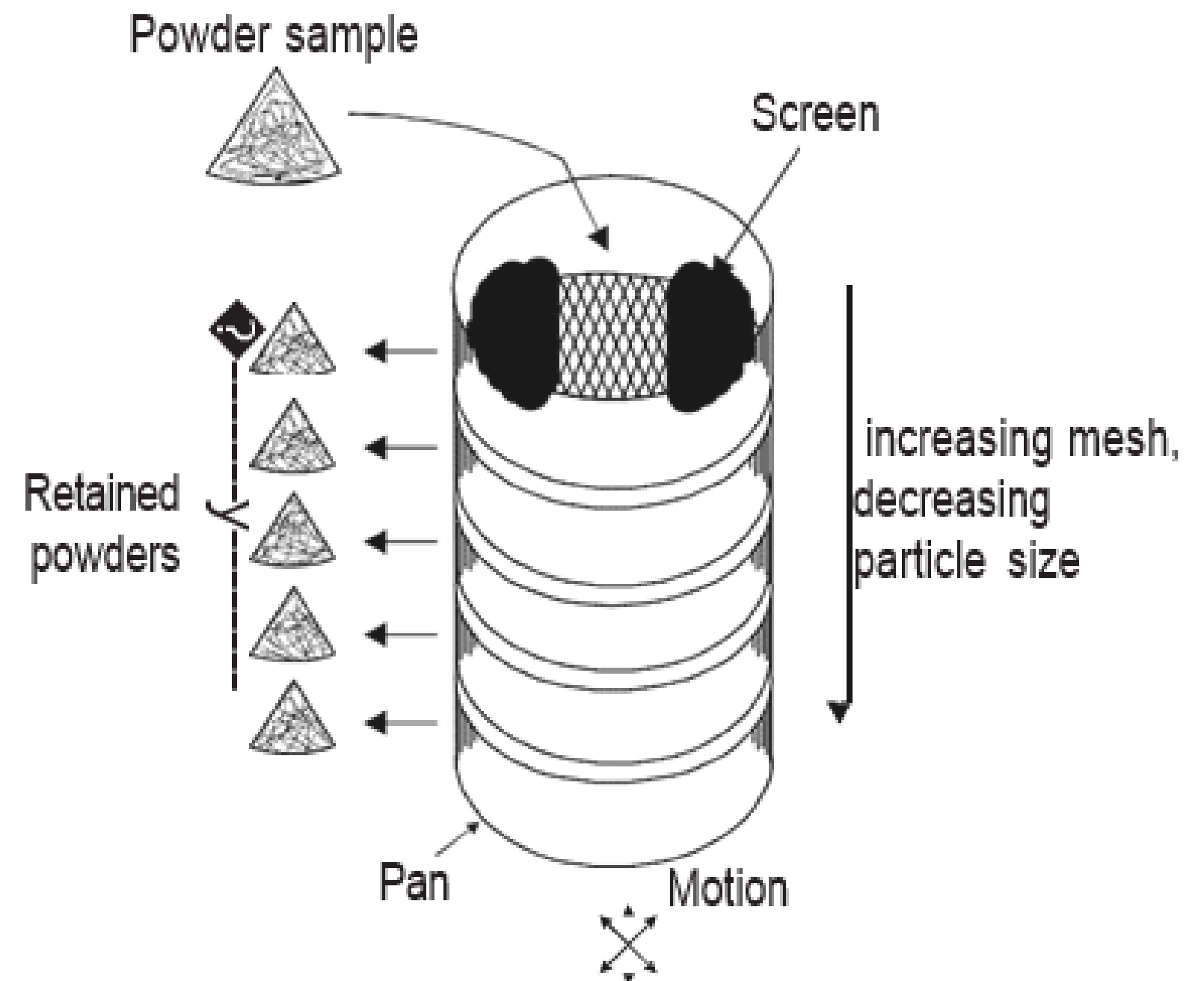
- Particle size relies on a stack of screens with decreasing openings that classify particles based on size.
- Woven-wire-cloth sieves made of copper, brass, bronze, nickel, Monel and stainless steel (and recently nylon) with square mesh pattern are used commonly.
- **Screens are designed in such a way that the size of the sieve openings vary in a geometric progression.**
- The standard step is an opening that changes by almost a constant factor $2^{1/4}$, indicating a step size of $2^{1/4}$ between each mesh.
- For example, the ratio of the 60-mesh opening size to the 70-mesh opening size (250–212 μm) is 1.18, the ratio of 170-mesh opening size to the 200-mesh opening size (90–75 μm) is 1.20 and the ratio of 325-mesh opening size is to 400-mesh opening size (45–38 μm) is 1.18.

- This means that each screen opening in a sieve analysis should be 19% nominally larger than the next screen.
- This amounts to a mass ratio of nearly 1.7 between successive classes.
- **Each sieve has a mesh number or mesh count which refers to the number of openings per linear inch of screen.**
- For example, a mesh count of 200 means there are 200 openings per linear inch.
- The mesh openings are square and, therefore, the count is the same in both directions.
- The total number of openings per square inch is $200 \times 200 = 40,000$.
- A large mesh means a small opening size and vice versa.

Standard Sieve Sizes (US)

Sieve designation mesh [ASTM]	Size	
	microns	Inch
31/2	5,660	0.218
4	4,760	0.1840
6	3,360	0.1300
8	2,380	0.092
12	1,680	0.065
16	1,190	0.046
20	840	0.032
30	600	0.0232
40	425	0.0164
50	300	0.0116
60	250	0.0097
80	180	0.0069
100	150	0.0058
140	106	0.0041
200	75	0.0029
230	63	0.0024
325	45	0.0017
400	37	0.00145

- In practice, a series of sieves are selected to cover the full range of particle sizes likely to be present in the powders.
- The sieves are stacked with largest mesh size at the top and a pan at the bottom



- An appropriate sample weight of metal powder (100 g) is spread on the top sieve which is then closed.
- The stack of sieves is agitated for a specified period of time by one or more of methods such as manual vibration, gyratory motion with tapping, mechanical shaking, mechanical-cum-electromagnetic vibration, accelerating air column or vacuum.
- **Care is taken to avoid clogging of the sieve openings by the powder particles.**
- The powder fractions remaining on each sieve and the bottom pan are weighed separately and reported as % retained or passed by each sieve.

- The powder passing through a particular mesh is designated by a minus (–) sign, and that retained on a mesh is designated by a plus (+) sign.
- For example, the designation of –120/+230 mesh for a powder indicates that it passed through a 120 mesh opening but not through a 230 mesh opening.
- The size of the powders would be in the range between 125 and 63 microns.
- Mesh sizes cannot go to very small opening sizes.
- **Sieve analysis is usually applied only to particles having sizes greater than 325 mesh or 44 μm .**
- Special sieves known as micromesh sieves having accurate opening down to 5.5 nm have been introduced for subsieve particle size analysis.

Microscopic Method

- Microscopy is the most definitive method of particle size analysis, as it involves actual counting of individual particles on a slide containing a sample of powder.
- **Though this is a tedious and time-consuming process, it is the most reliable and standard method for checking the accuracy of other methods.**
- Optical microscope is used for particles ranging from 100 to 0.1 μm .
- Scanning Electron Microscope (SEM) is used for particles ranging from 10 μm to 0.01 μm while Transmission Electron Microscope (TEM) is used for particles in the range 10 μm to 0.0001 μm .
- Observation of the powder particles by SEM and TEM, though may not be used for quantitative purposes, is generally very useful since these allow examination of individual particles which can reveal features not given by other methods.

Sedimentation

- Sedimentation analysis is one of the most widely used particle sizing methods along with specific surface area method.
- **Sedimentation is used for classifying metal powders based on their rate of settling in a fluid medium.**
- This measurement technique is based on **Stoke's law** of fluid dynamics, which states that at low velocities, the frictional force on a spherical body moving through a fluid at constant velocity is proportional to the product of the velocity, the fluid viscosity and the radius of the sphere.
- The settling velocity of falling spherical particles at low velocity in a fluid of infinite extent is given by

- Terminal velocity (V) is given as
$$V = \frac{(\rho_p - \rho_l)dg^2}{18\eta}$$

where

- ρ_p is the density of the particle in g/cc
- ρ_l is the density of the fluid in g/cc
- η is the viscosity of the fluid in poise
- g is the acceleration due to gravity in cm/s²
- d is the diameter of particle in cm

- Sedimentation involves the suspension of the powder sample in a fluid medium and allowing it to settle for a suitable time and measuring the settling velocity in the suspension.
- The diameter of a powder particle D , settling under gravity and assumed to be a sphere is given by

Where

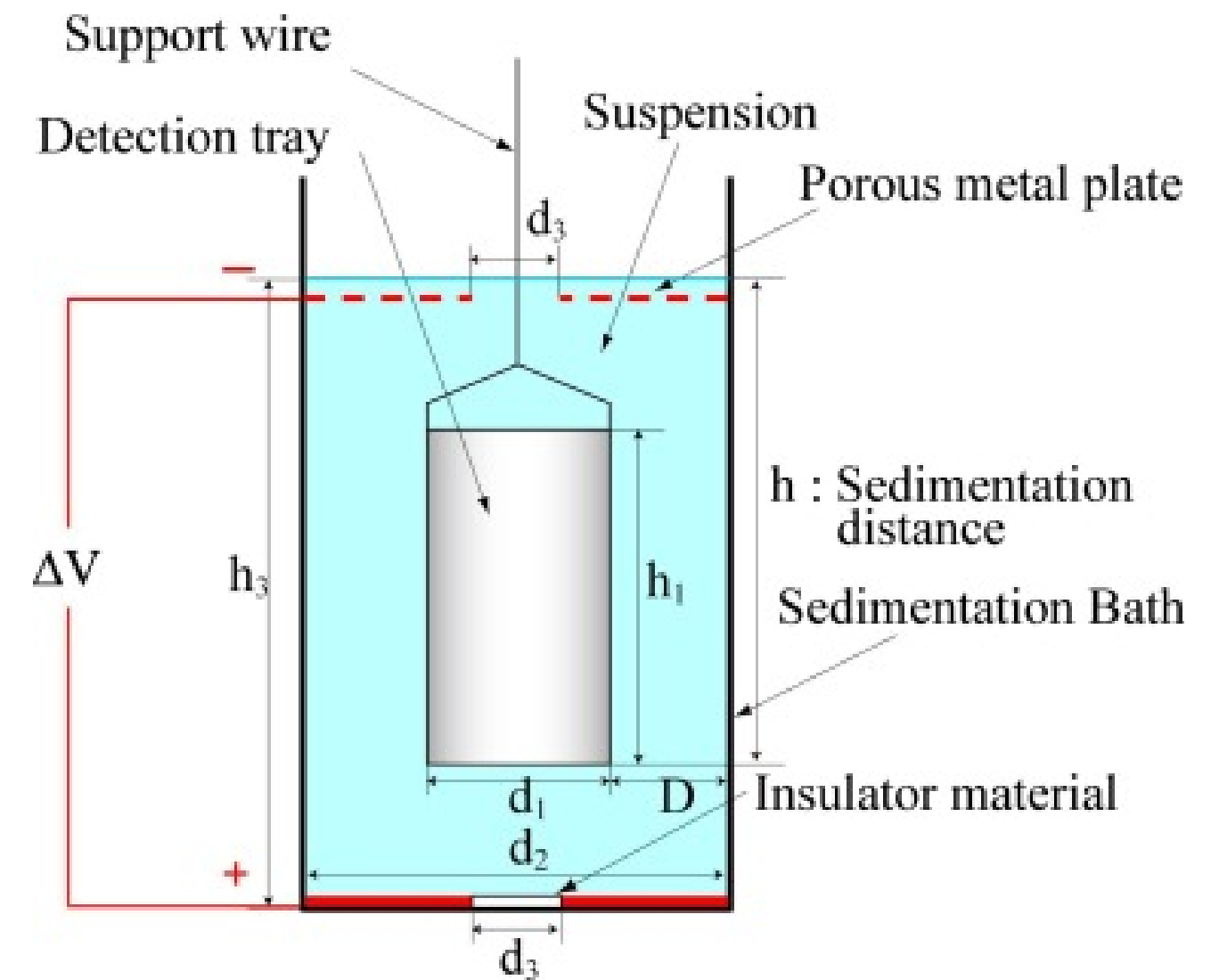
h is the viscosity of the liquid, d_r is the difference in density between the liquid and powder, and g is the acceleration due to gravity in cm/s^2 .

$$D = \left[\frac{18\eta v}{d_r g} \right]^{1/2}$$

Settling velocity in cm/s , d_r is the difference in density between the liquid and powder, and g is the acceleration due to gravity in cm/s^2 .

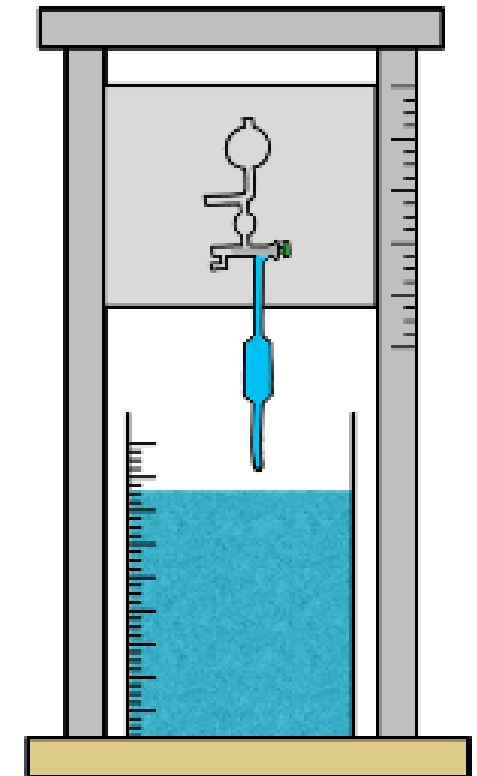
Sedimentation Balance

- One pan of a balance is immersed at a known depth in a liquid column containing the metal powder in suspension.
- The suspended pan with the suspension is balanced first and the increase in weight due to settling of some of the particles on the balance pan is recorded at intervals and a sedimentation curve is obtained.
- An advantage of this method is the direct weighing of the sediments with accuracy and without disturbing the suspension, so that a precision of 0.1% can be obtained.



Pipette Method

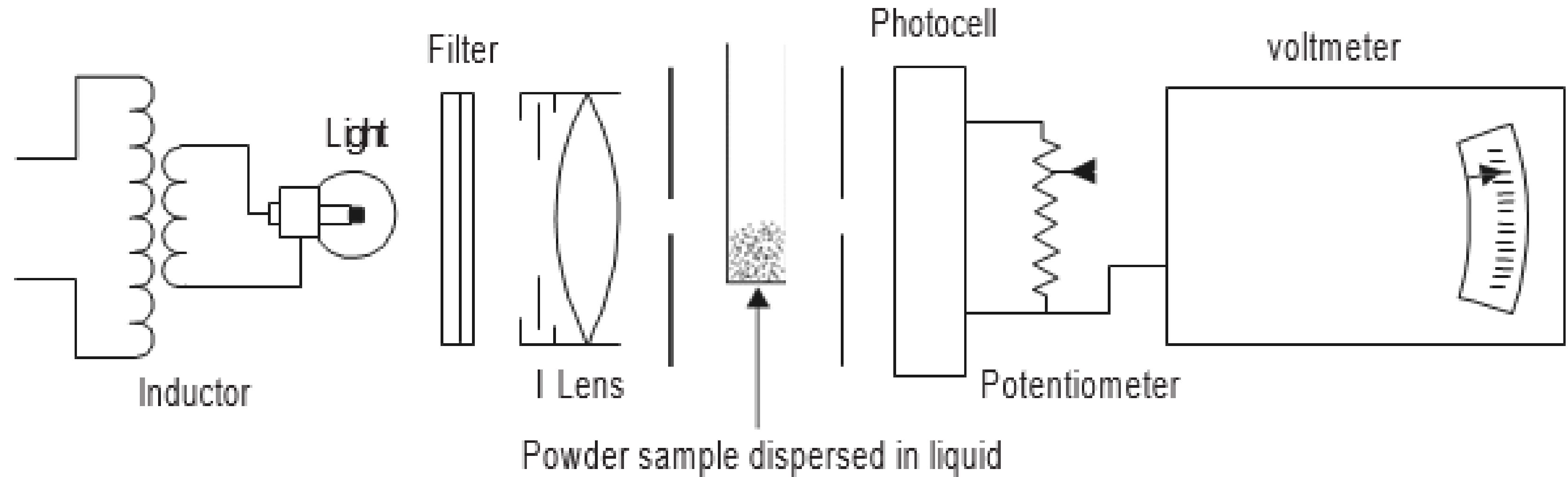
- In this method, the suspension is allowed to settle in a cylindrical vessel and withdrawing the suspension from a certain depth, after different time intervals, drying and weighing.
- This is relatively quick, reliable and inexpensive.
- In the hydrometer or specific gravity method, the concentration of the sample taken in a pipette is determined by measuring the specific gravity of the suspension at a specified depth and at certain time intervals using a hydrometer.
- A sedimentation curve is plotted from the results.
- **This method eliminates the necessity for drying.**



Turbidimetric Method

- In this method, a sample of powder dispersed in a liquid is poured into a glass cell and allowed to settle.
- A collimated beam of light is passed through the cell.
- The variation of the intensity of transmitted light through the dilute suspension contained in the cell at a certain depth as function of time is recorded by photocell and millivoltmeter.

Schematic diagram of turbidimeter





- The intensity of the liquid beam passing through the clear solution is adjusted to 100% and the concentration of the suspension is adjusted so that the reading for the intensity of the light beam through the suspension before any settling takes place, is in the range of 20–40% of intensity through the clear solution.
- As the suspension settles, the projected area of the particles in the suspension decreases and the intensity of the light beam increases.
- Initially, all the particles are distributed uniformly through the volume of the sedimentation cell.
- As settling proceeds, the larger particles settle faster than the smaller ones.

- The time **t_x needed for a particle of size x** to settle through the distance h from the top of the suspension to the level of the light beam can be calculated

$$t_x = \frac{18\eta h}{(\rho_p - \rho_l)gx^2}$$

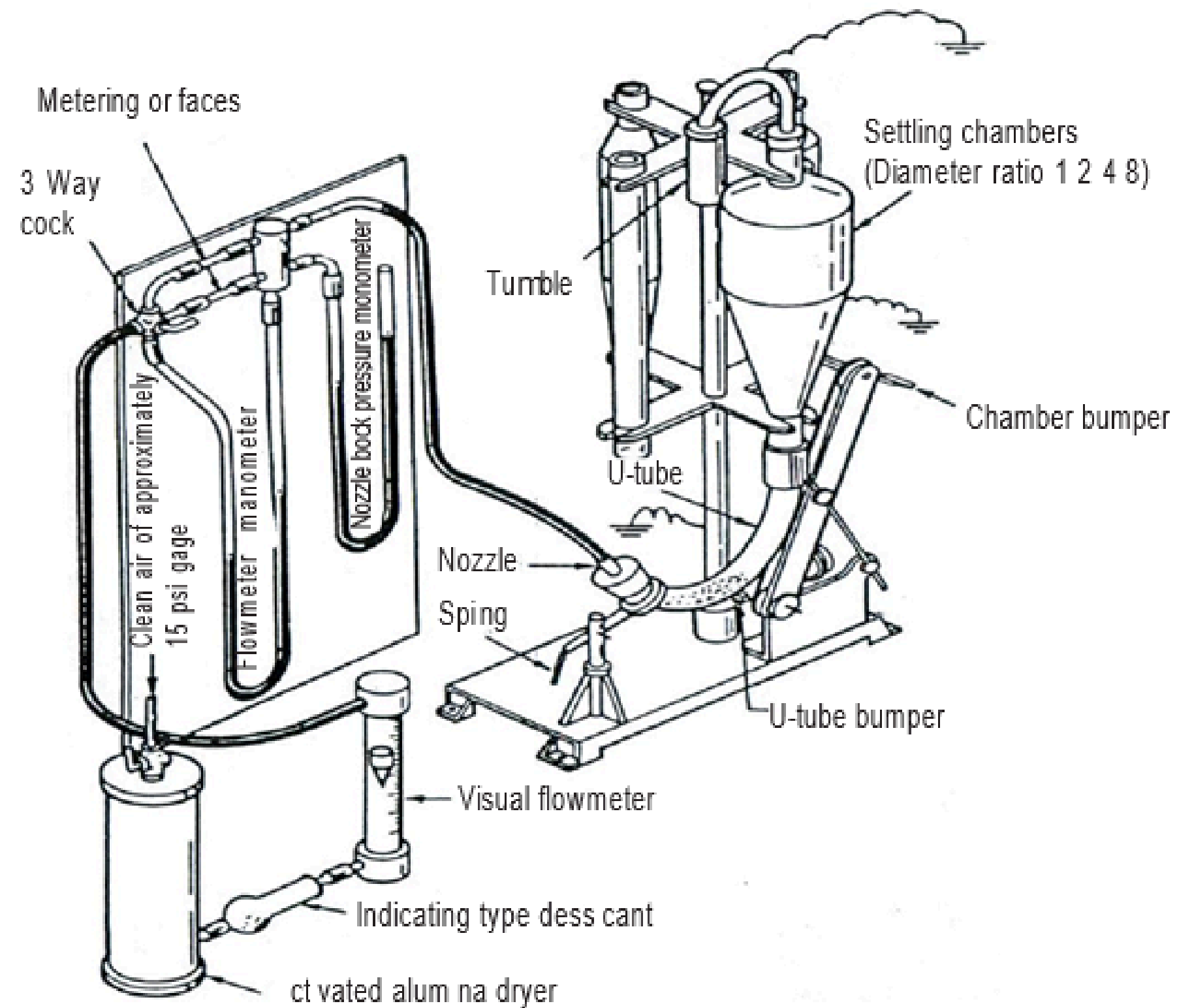
- Where, h is the viscosity of the liquid, ρ_p and ρ_l are the densities of the particles and the liquid respectively and g is the acceleration due to gravity

- X-rays can be used instead of a light beam.
- **However, in the case of X-rays, the attenuation of the X-ray intensity will be proportional to the mass of the powder particles instead of the projected area.**
- This method is used to determine the size distribution of spherical and non-spherical metal particles.
- X-ray method, though not accurate, is useful for comparing the fineness of same metal powder prepared under different conditions.
- The method is quick and requires a very small amount of powder.
- In recent years, laser beams have also been successfully used in place of light or X-ray beam.

Elutriation

- Elutriation method also relies on the application of **Stoke's law**.
- This method is based on the principle that **particles with settling velocity less than the velocity of vertically raising fluid (liquid/gas) will be carried upwards**.
- Those particles having higher settling rates will fall in the column (i.e. heavier particles will come down).
- **The elutriation differs from sedimentation in that the fluid is in motion in upward direction against the falling particle.**
- This method is used for subsieve particles and determines the size fractions of powder particles.
- This method can be used to classify the powders and to remove unwanted fines.
- The method can handle large samples compared with sedimentation and decantation method.

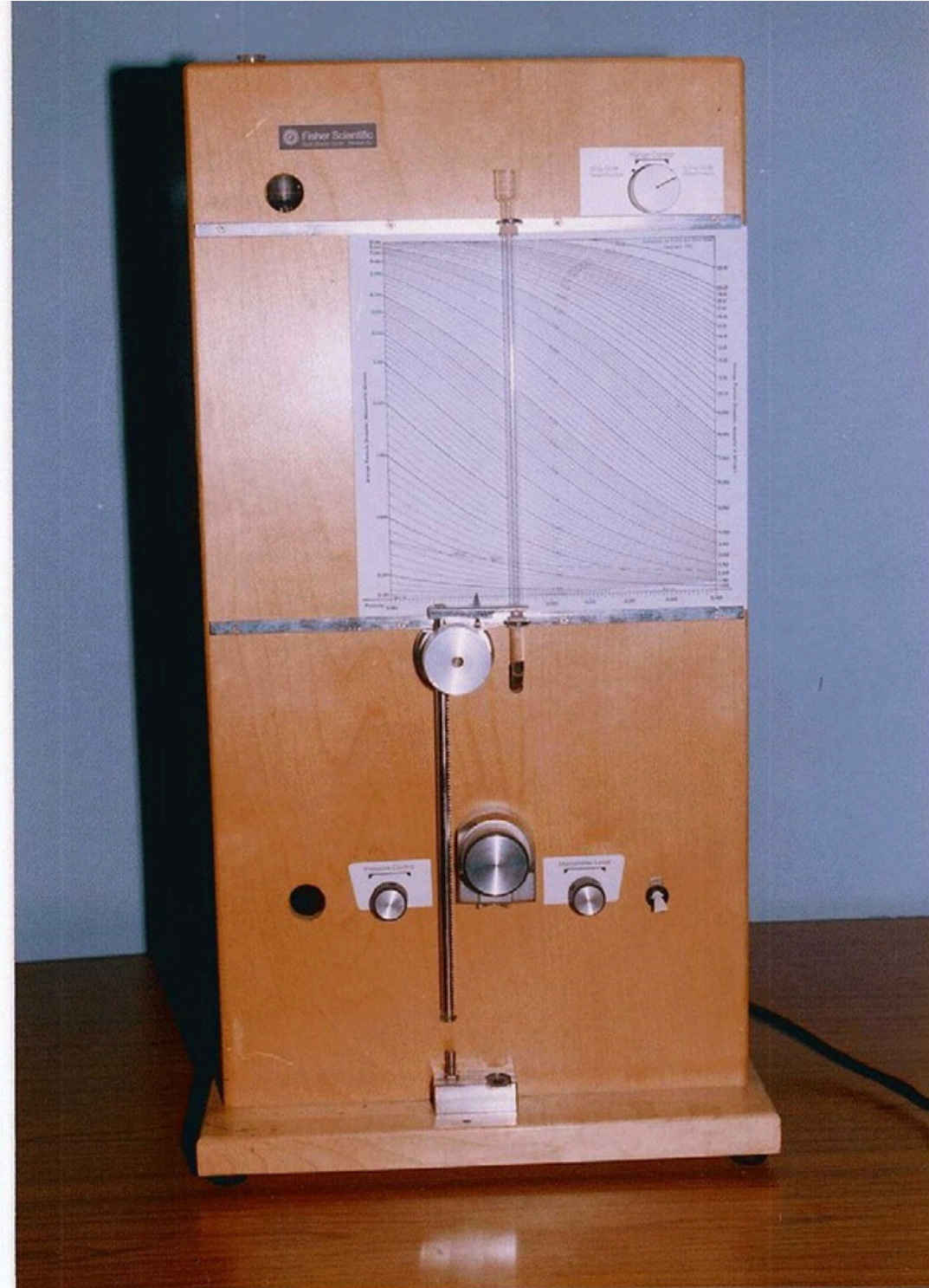
Roller Air Analysers

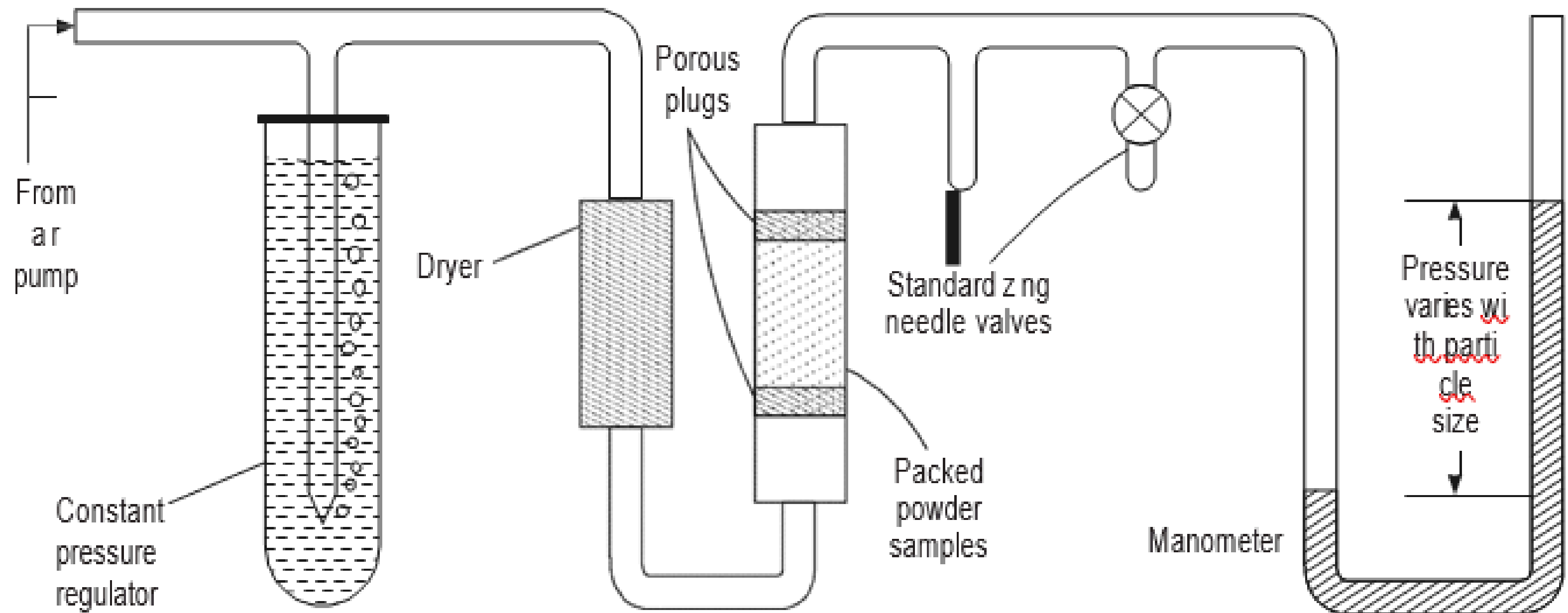


- Determine the particle size distribution of metal powders by suspension in a stream of air.
- A stream of high velocity air flowing through a nozzle of suitable size impinges on the powder sample contained in a U-tube, so that the powders become dispersed within the stream.
- The suspension of the powders in air rises through a cylindrical settling chamber.
- The velocity of the air stream through the chamber, balancing the settling velocity of the particle as given by Stoke's law is
- $V = 29.9 \times 10^{-4} d^2$
- V is the velocity of air stream in cm/s
- ρ_p is the particle density in g/cm³
- d is the particle diameter in micron

- Particles having sizes smaller than d for a given air velocity will be carried through the settling chamber into the collection system to an extraction thimble.
- Larger particles fall back into the U tube.
- By using a series of vertical settling chambers, with diameters in the ratio 1:2,4:8 etc, and by maintaining a constant volumetric flow rate, the powders can be classified into different particle size fractions.
- This method can classify particle size fractions with the maximum sizes of particle in the ratio of 1:2, 4:8 (5 to 40 microns).

Fischer Sub sieve Sizer





- Fischer subsieve sizer measures the flow of air through a packed powder bed.
- From this measurement, the average particle size (and porosity) can be calculated using a chart, which is an integral part of the instrument.
- The Fischer subsieve analyzer is used extensively for process control.
- It measures accurately and precisely the permeability.
- **However, it does not measure the surface area or particle size directly.**
- The flow through a porous compact depends on the structure of the pores. Flow, therefore, depends on the porosity, particle shapes, particle size and particle size distribution as well as the method of compaction, all of which affect pore structure.
- Conversion of the measured permeability to surface area is carried out using the **Kozeny-Carman equation**

$$K = E^3 (1.0 - E^2) B S_w^2 \rho^2$$

where

$$E = (V_{\text{bed}} - V_{\text{powder}}) / V_{\text{bed}}$$

$$K = \text{permeability of the powder bed}$$

$$E = \text{porosity of powder bed}$$

$$V = \text{volume (cm}^3\text{)}$$

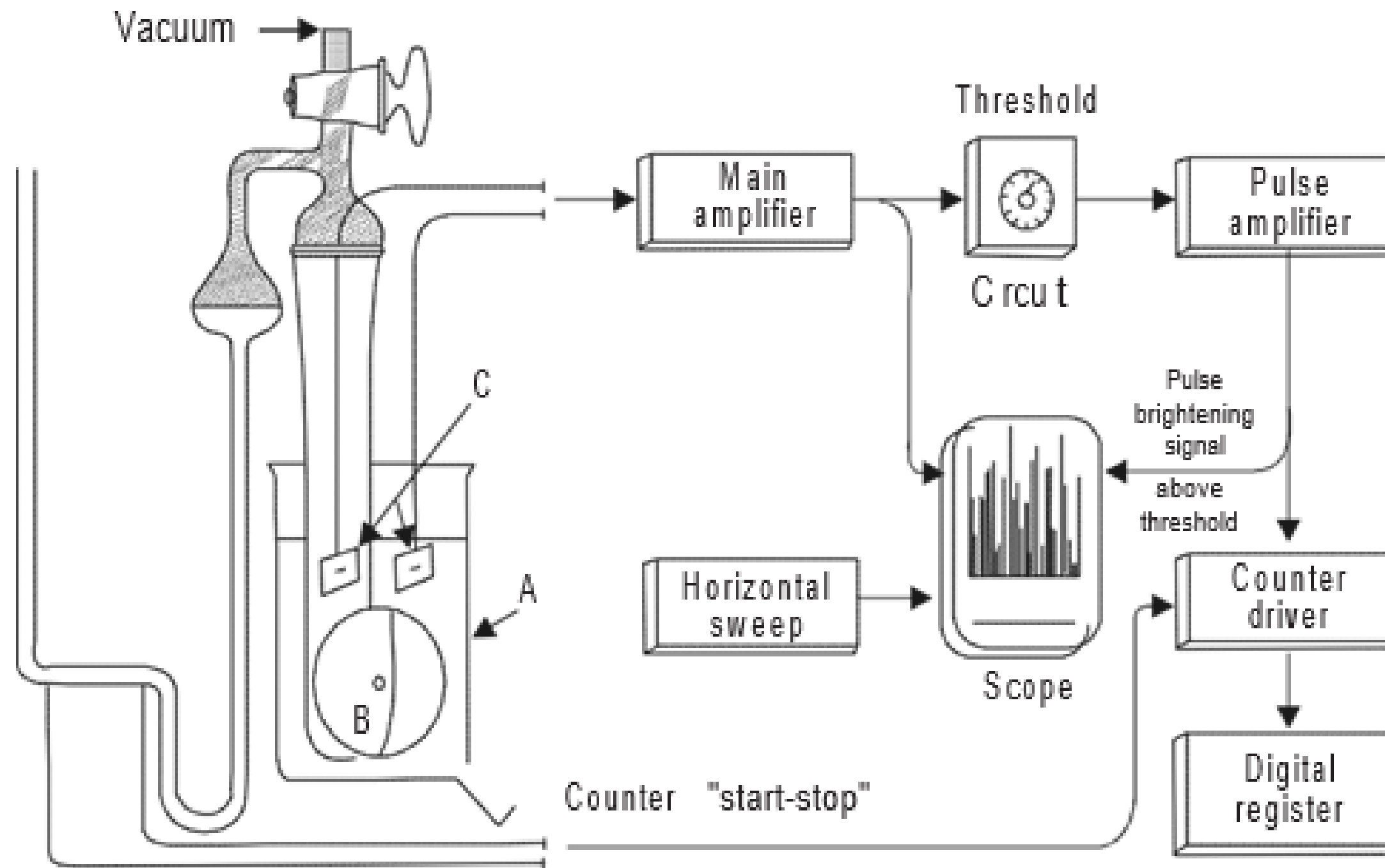
$$\rho = \text{density of powder material (true) (g/cm}^3\text{)}$$

$$S_w = \text{surface area per unit weight of powder (cm}^2\text{/g)}$$

$$B = \text{a constant}$$

- From the above equation, it can be seen that permeability is related to bed porosity, surface area and particle density.
- Average particle diameter (d), also known as the Sauter mean diameter, is related to the surface area by the relation $d = 6/r Sw$.
- The average particle size is calculated using the calculator chart provided with the instrument.

Coulter Counter



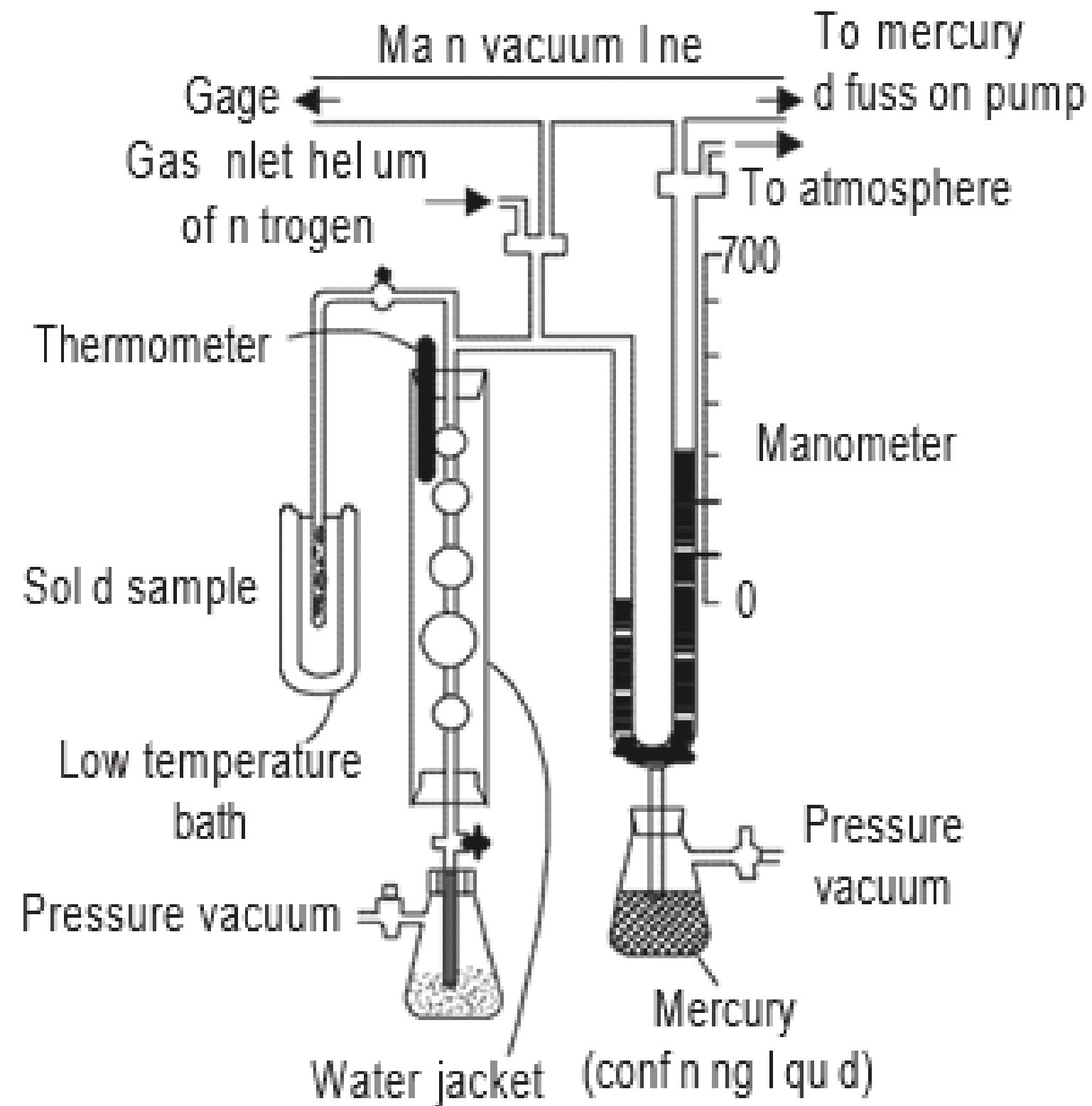
- Powder sample is suspended in a conducting liquid.
- The suspension is circulated through a sampling chamber, which contains a sensing orifice with immersed electrodes on either side.
- The passage of the particles through the sensing zone changes the resistance of an electric circuit through the liquid between the electrodes.
- This produces a voltage pulse proportional to the particle volume.
- When the powder is streaming pass the electrode, it replaces an equal volume of electrolyte (Archimedes principle) causing a change in resistivity.
- The voltage pulse is independent of particle shape and hence diameter is expressed as that of the sphere of equal volume or equivalent spherical diameter.

- The unit is capable of detecting the volumes of individual particles in a liquid suspension from tens to thousands of particles per second.
- This method, also known as **electrozone** method is capable of measuring particle diameters ranging **from 0.1 to 2,000 mm.**
- Electrolytes made of conducting aqueous solutions of common salts such as sodium or potassium phosphates, chlorides, sulphates or nitrates can be used.

SURFACE AREA (SPECIFIC SURFACE)

- Specific surface area or specific surface is defined as the total surface area per unit weight
- It depends on size, shape, density and surface condition of the particles.
- The **main methods** used to determine the surface area are:
 - *Gas adsorption method* determines the surface area of a powder by measuring the amount of gas adsorbed by a monomolecular layer of the powder.
 - *Permeametry* measures the resistance to fluid flow through a compacted powder bed.

Gas Adsorption Method



- Gas adsorption starts with a clean powder surface achieved through a vacuum bake out.
- This clean surface is exposed to an adsorbing gas, most typically by chilling the powder to liquid nitrogen temperatures (77 K or -196°C or -384°F) when nitrogen condenses on the powder surface.
- The larger the surface area, the greater the amount of adsorbed nitrogen.
- The measurement gives the BET specific surface area, named after Brunauer, Emmett and Teller, who developed this method.
- These instruments are highly automated, but the measurement is usually applied only to subsieve powders.

- The relationship between specific surface area S_w , the average particle diameter d on an equivalent sphere basis and the density of the material ρ , can be derived as follows:

In case of irregularly-shaped particles, the surface area will be generally high. For calculating the specific surface area, a correction coefficient K is employed.

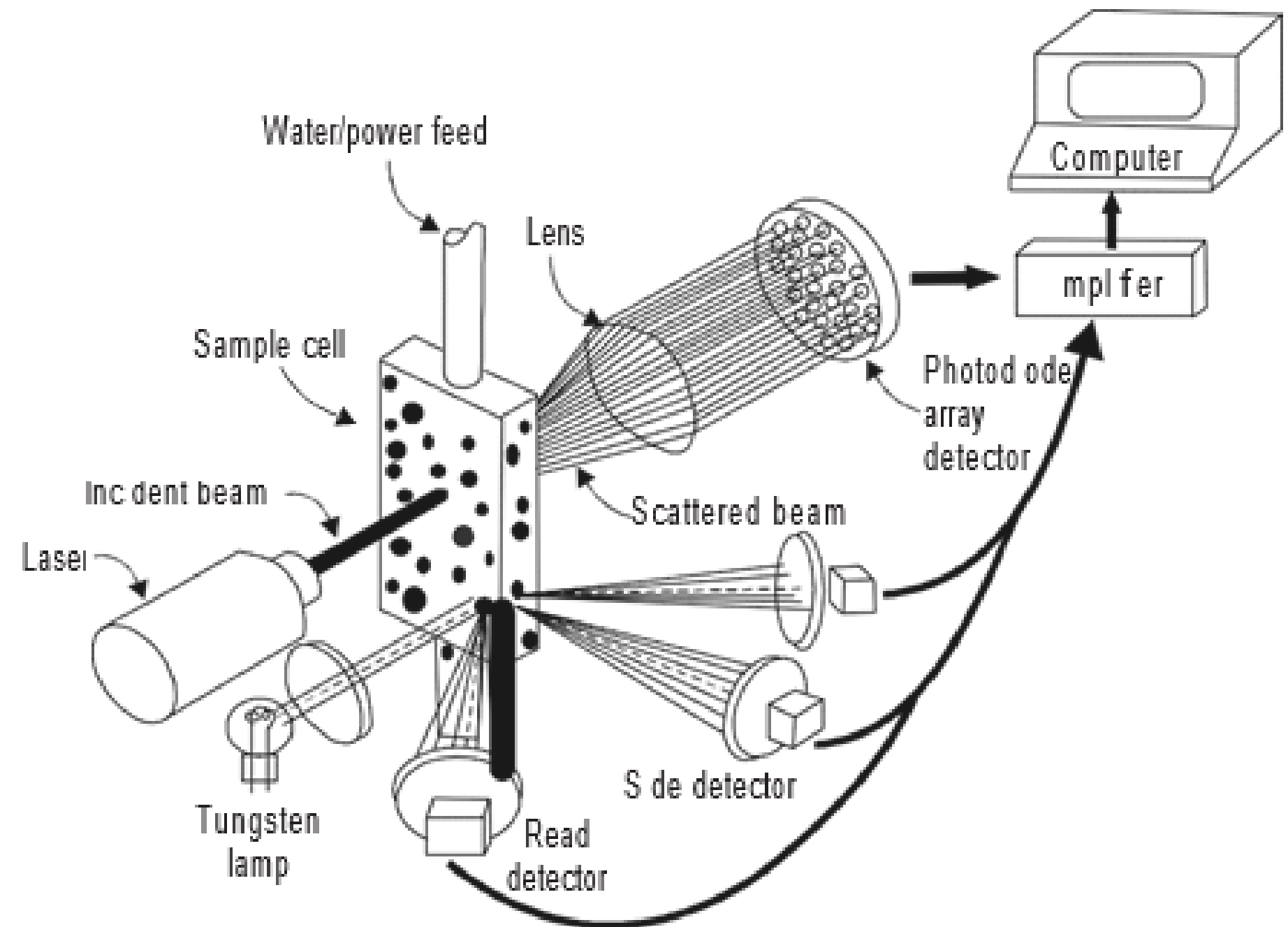
$$S_w = \frac{6}{d\rho}$$

K varies from 2 for rounded powders to about 10 for dendritic powders

LASER LIGHT SCATTERING

- Light incident on small particles is scattered through angles that are inversely proportional to particle size.
- Scanning of the angular distribution of scattered light flux provides information of particle size and size distribution.
- The system consists of optical, electronic and sample handling units.
- The powder sample is suspended in liquid medium and placed in the sample zone.
- This zone is then illuminated by a helium-neon light.
- The particles cause the light to be scattered, which is collected by a lens and directed towards an optical filter.

- The transmitted light from the filter is focused on to a photodetector, which produces electrical signals.
- **These electrical signals are converted to digital signals and used to compute particle size distribution.**



MICROSTRUCTURAL FEATURES

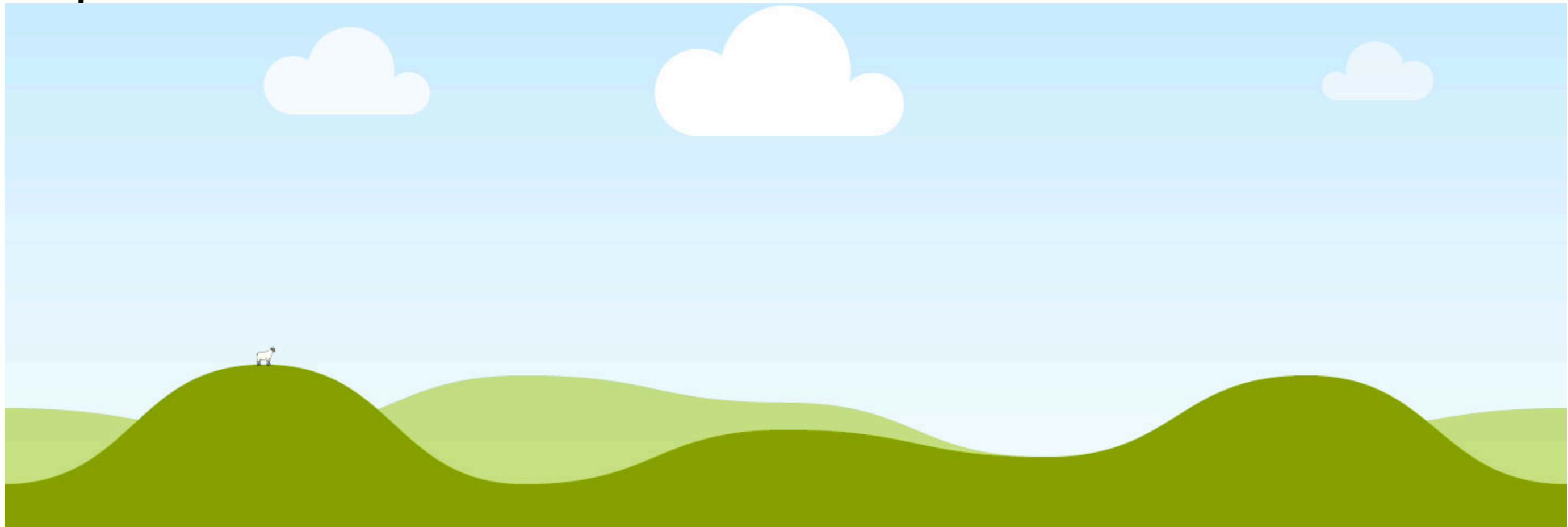
- Microstructural examination also provides information on the deformation of powders produced by mechanical methods like milling which may be an added advantage during sintering, increasing the driving force for sintering.
- Grain size of powders can be studied using microscopy.
- In general, the grain size of a powder is dependent upon the method of production.
- Fine grain size gives better sintering and uniform shrinkage.
- Surface texture is another characteristic, which can be studied using microscopy.

PACKING AND FLOW CHARACTERISTICS OF POWDERS

- Small powders or irregularly-shaped particles do not flow or pack easily.
- Sharp corners are a source of friction between particles and inhibit flow and packing.
- Both properties are important to powder shaping, since the powder needs to fill out a die cavity rapidly and uniformly.
- The packing density decreases because of poor flow of particles.

Angle of Repose

- It is an index of friction.
- It is the angle from the horizontal observed by pouring a powder into a pile.



- Angle of repose is of practical interest and is closely related to interparticle friction and the flowability.
- Larger angles indicate greater interparticle friction.
- **Four different methods are used to measure the angle of repose.**
 - i. **Fixed height cone:** The powder is carefully poured through a funnel at a fixed height until apex of the heap formed by the powder reaches the bottom of the funnel.
The tangent of the angle of the repose is the ratio of the height to the mean radius of the base of the powder heap.
 - (ii) **Fixed base cone:** The powder is allowed to flow through a funnel which is raised vertically until the heap covers a circular base of fixed size.
The tangent of the angle of repose is calculated in the same manner as the fixed height cone method.

i. Tilting table: The rectangular box filled with powder is tilted until the contents begin to slide.

The angle of the upper surface of the box makes with the horizontal is equal to the angle of repose.

(iv) Rotating cylinder: A sealed hollow cylinder half full of powder is rotated until the surface of the powder exhibits its maximum angle with the horizontal, which gives the angle of repose.

Angle of repose is influenced by powder characteristics like particle size, shape and cohesiveness.

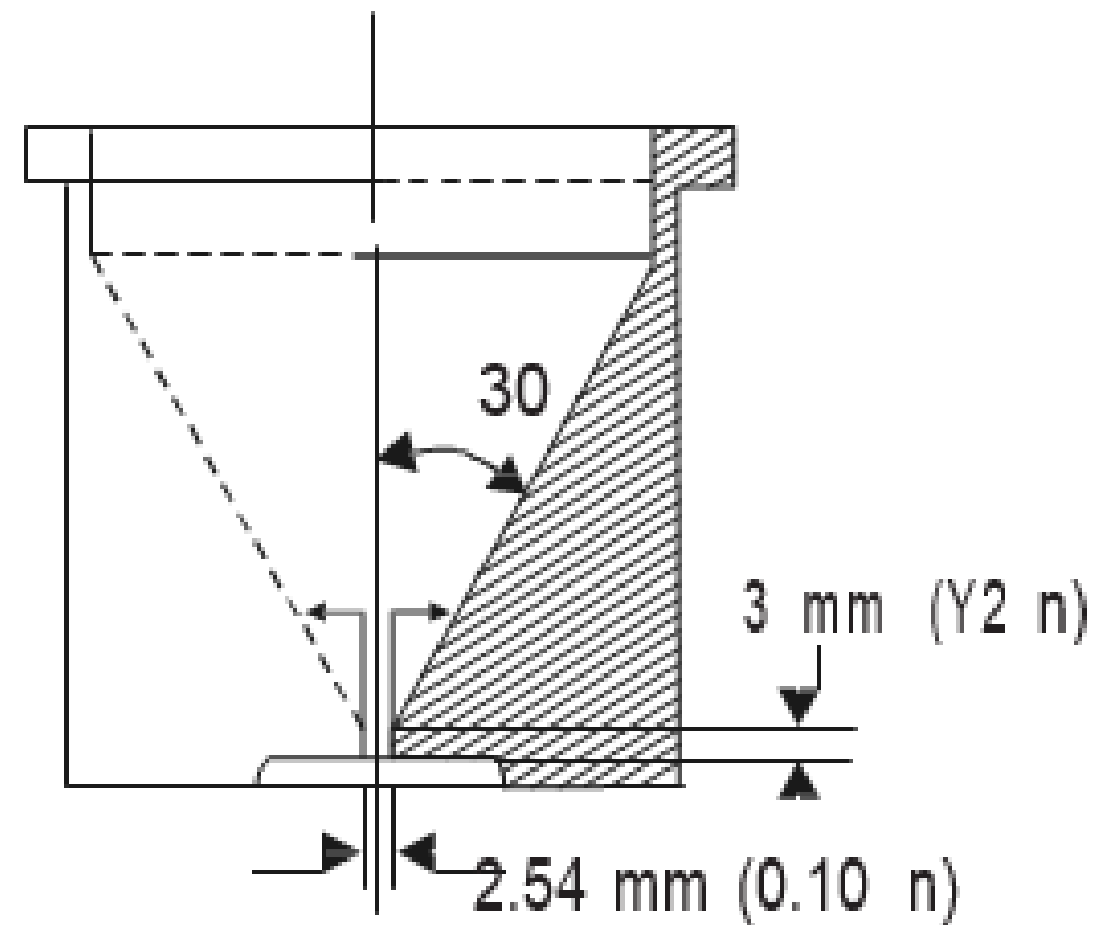
A high angle of repose indicates considerable difficulty for particles to flow past one another.

Spherical particles have smaller angle of repose than irregularly-shaped particles

Flow Rate

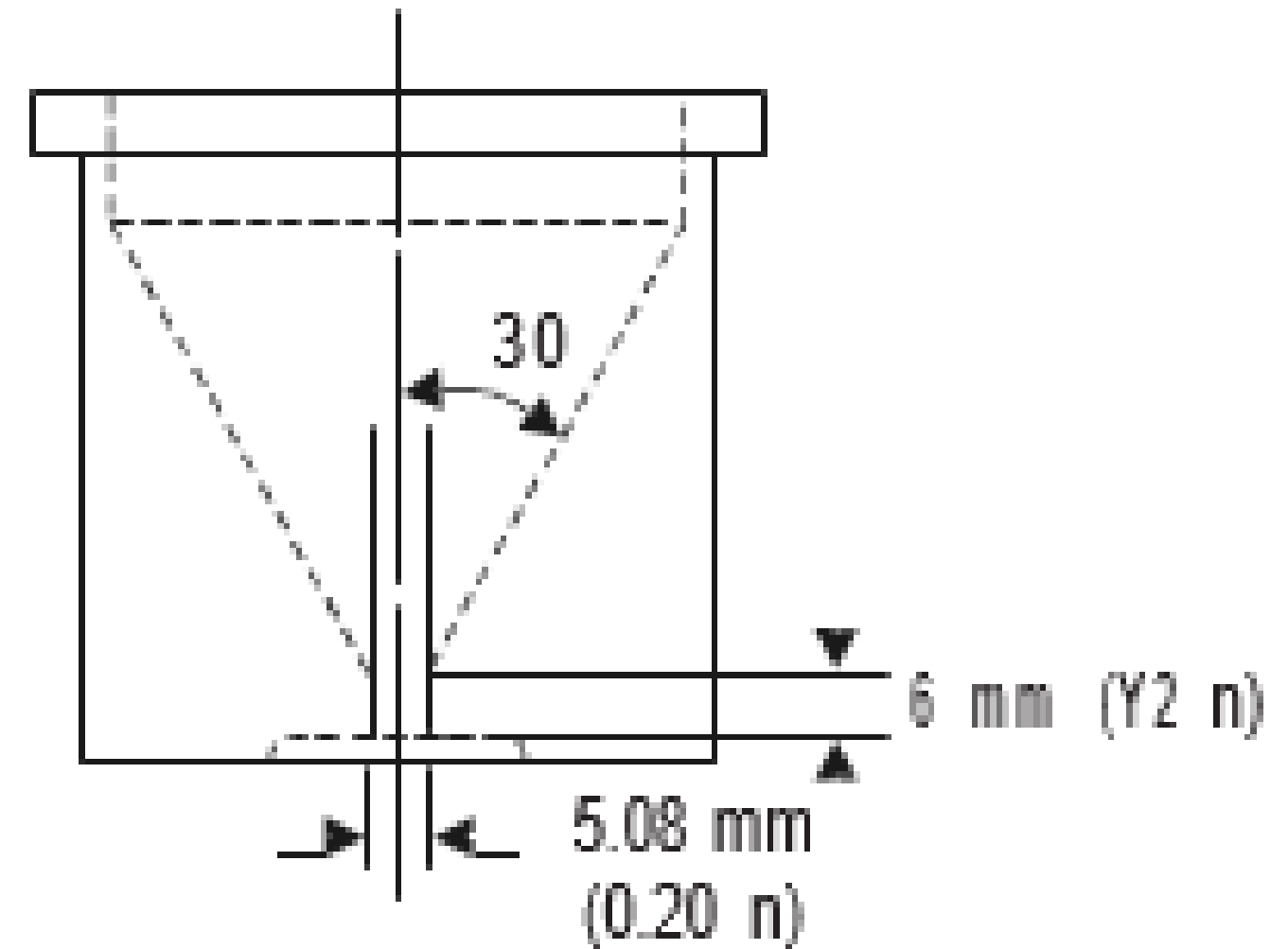
- Ease of flow of powders is the flow time, measured by the time taken for a known weight of powder sample to fall through a specially designed funnel called the Hall funnel.
- The apparent density and flow times are easily obtained with the Hall flowmeter by combining a precision volume cup with the funnel.
- The apparent density is the weight of powder divided by the cup volume.

Hall flowmeter

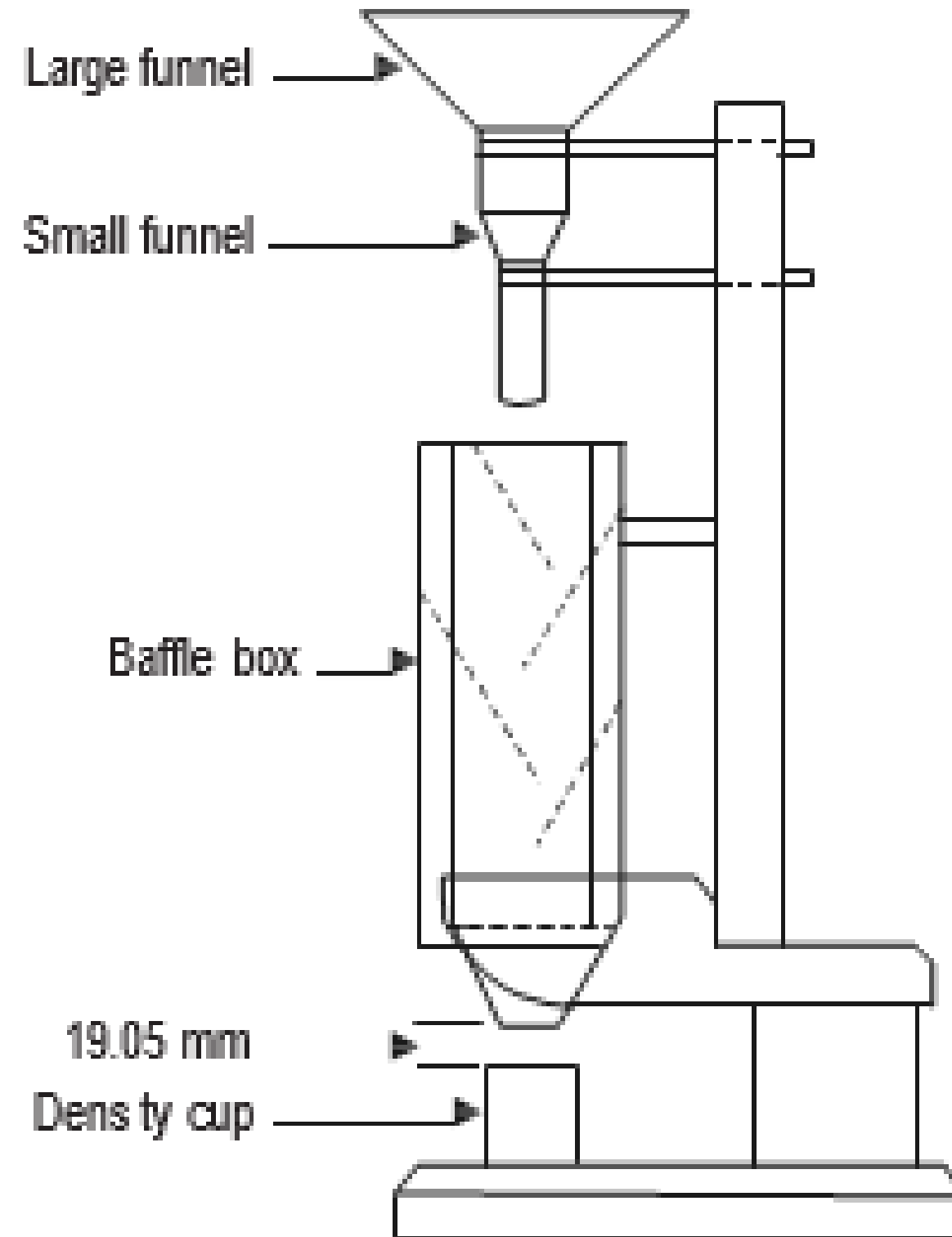


- Fifty grams of dry powder sample is transferred to the cup through the funnel and the time taken for the powder to be transferred to the cup is found out by using a stopwatch.
- The flow rate is reported as the time in seconds taken for 50 grams of powder to flow through the orifice of the funnel.
- Flow characteristics depend on interparticle friction, particle shape and size as well as the type of material.
- **Short flow times indicate free-flowing powders while long times suggest high interparticle friction.**
- Poor flow characteristics cause slow and non-uniform press feeding and difficulty in ensuring even fills of the die cavity.

- Small, subsieve powders do not flow easily and are designated as non-free-flowing.
- In the case of non-free-flowing powders like tungsten, another funnel called the **Carney funnel** is used.
- This funnel has an opening twice that of the Hall flowmeter funnel.



- **Scott volumeter** is another apparatus used for measuring flow rate of non-free-flowing powders like tungsten carbide.



DENSITY

- Measurement of density permits the distinction between different materials as well as the determination of inaccessible porosity of powder or a finished P/M part.
- The technique used is **pycnometry**, which determines the density by measuring the difference between the specific and bulk volumes of a sample.
- This method is based on the displacement principle using the powder as the solid and dry helium as displaced medium.
- **The method can be used to determine total pore volume, but it does not provide quantification of pore size or distribution.**

Apparent Density of the Metal Powder

- Mass per unit volume of loose or unpacked powder expressed as g/cm^3 .
- Apparent density depends on the density of the solid material, particle size, particle size distribution, particle shape, surface area, roughness of the individual particle and particle arrangement.
- Apparent density decreases with decreasing particle size, decreases as the particle shape becomes less spherical and more irregular, and also decreases with increasing surface roughness.

- **Hall flowmeter. (Used for finding flow rate)**
- Apparent density measurements are made by pouring powder into the funnel and allowing it to flow into the density cup.
- The volume of density cup is 25 cm³.
- After the cup is filled the funnel is moved away and the excess powder is carefully levelled off using a spatula (or) straight edge.
- Care must be taken to avoid physical densification of the powder in the cup when levelled.
- The apparent density is then determined by weighing the powder in the cup in grams and dividing by the volume of the cup.
- **For non-free flowing powders, the Carney funnel which has an orifice diameter twice that of the Hall funnel is used.**

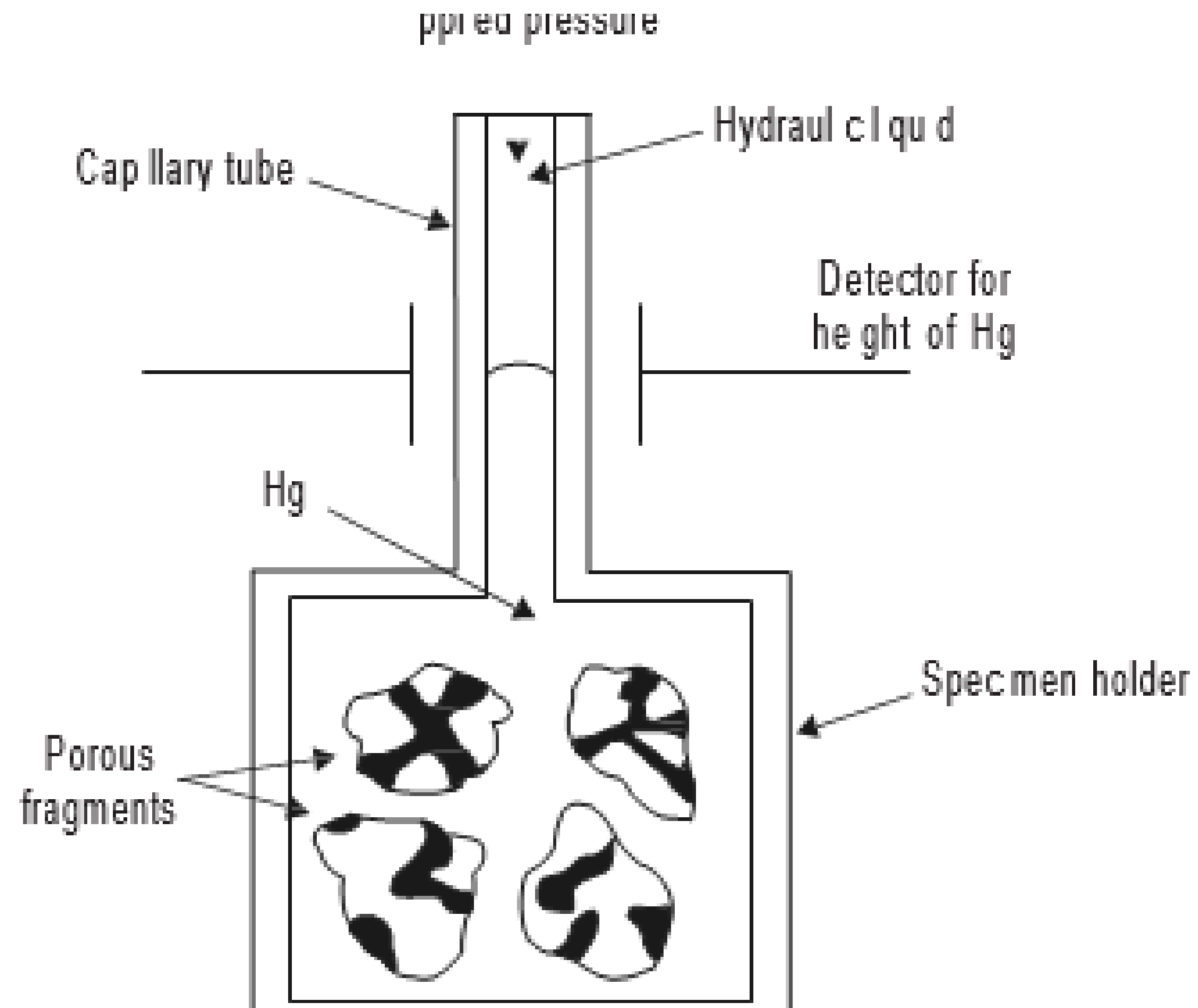
Tap Density

- Tap density is the density of the powder, after it has been mechanically shaken down or tapped until the level of the powder no longer falls.
- **Tap density is always higher than apparent density.**
- To determine tap density a standard weight of powder, usually 5 gram is weighed accurately.
- The powder is poured into a clean dry graduated cylinder.
- **The powder is tapped and allowed to settle in the cylinder by mechanical or manual means until no further decrease in volume is observed.**
- In the case of mechanical tapping, a tapping machine capable of delivering 100–250 taps per minute without loosening of surface layers is employed.
- The volume of the fully densified powder sample in a graduated cylinder is read and used to calculate tap density.

POROSITY

- Porosity is expressed as a percentage which is equal to 100% observed density.
- % Porosity = 100 - % theoretical density
- **Mercury porosimeter** measures the volume of mercury intruded into the pores of a powder sample as a function of pressure applied to the mercury.
- This method is used for size and size distribution measurement over a wide range ($5 \times 10^6 \text{\AA}$ to $30 \text{\AA}$) depending upon the capacity of the apparatus used.
- The porous material to be analyzed is placed into the mercury porosimeter and the volume of mercury absorbed by the pores is measured as a function of the pressure applied to the mercury.

Mercury porosimeter



COMPRESSIBILITY OF METAL POWDER

- Compressibility and compactability are terms used to describe the **extent to which a mass of powder can be densified by the application of pressure alone.**
- Compression ratio, a related term is the ratio of the final pressed density to the apparent density of the powder.
- This ratio determines the depth of the die cavity required to hold all the powder needed to meet the specified part density.
- Powders of high apparent density are preferred because tooling can be made shorter, and thus, stronger.
- **The compressibility of a powder is influenced by several factors.**

- i. Inherent hardness of the metal or alloy:** The powder particles undergo considerable deformation during pressing and hence their work hardening characteristic has a strong influence on compressibility.
- ii. Particle shape:** The more irregular the powder lower is its compressibility.
- iii. Internal porosity:** Fine internal porosity in a powder tends to become isolated during pressing. This traps air within the particles. Although air is highly compressible, it takes up volume and contributes little to weight. Non-porous powders have the highest compressibility.

- i. **Particle size distribution:** Powders of a uniform size exhibit relatively poor compressibility. Mixtures of particles sizes designed to occupy the most space have better compressibility.
- ii. **Presence of nonmetallics** such as unreduced oxides reduces compressibility because of their hardness and low specific gravity.
- iii. **Use of solid lubricant:** Lubricants added to aid the pressing of metal powder affect compressibility because of their low density and the high volume they occupy. Addition of elements such as **graphite and sulphur** generally degrade compressibility.

- Typically, cylindrical or rectangle test sample are prepared by pressing the powders in a die with pressure applied simultaneously from top and bottom.
- The pressure required to achieve a specified density is the measure of compressibility.
- The compressibility can also be specified on the basis of density achievable at a given pressure.
- By plotting the density obtained by a series of increasing levels of pressure, compressibility curve for specific powders can be obtained.

- Sufficient powder to form the test bar (25 mm diameter; height 12.7 mm to 24 mm) is loaded into the appropriate die and a pressing load is applied to the opposing punches.
- When the specified pressure is reached, the part is ejected from the die and its height and peripheral dimensions are measured and the volume is determined.
- The weight of the part divided by the volume is a measure of the density achieved at the specified pressure.
- Compressibility is reported as the density in grams per cubic centimeter at a specified compacting pressure.

STRENGTH PROPERTIES

- Strength of both green compact and sintered parts are generally measured.
- Green strength of compacts is measured for evaluating the ability of the compact to withstand handling during subsequent processing.
- Strength properties of sintered parts are important from the service point of view.

Green Strength

- It is the mechanical strength of an unsintered powder compact.
- Strength of green compact results mainly from interlocking of irregularities on the particle surface, which is promoted by plastic deformation during pressing.
- The **Rattler test** involves falling of compacts of 1/2" diameter and 1/4" thick at 87 revolutions per minute for 1,000 revolutions in a cylindrical barrel of 14 mesh bronze screen (3 x 3/4" diameter, 4" long).
- The loss in weight of specimen expressed as a % of original weight is an inverse measure of green strength (S).

- The strength of green compact can also be determined by **transverse bend test**.
- A rectangular specimen of 1/2" wide, 1/4" thick and 1¼" long held between two supports 1" apart.
- A uniformly increasing load is applied on the flat surface of the compact at the center until the specimen breaks.

$$S = \frac{3WL}{2BT^2}$$

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- S is the green strength in lb/in², W is the l between the supporting rods, B and T are specimen expressed in inches

- Powders with irregularly-shaped particle produce compacts of lower green strength than powders with spherical shape.
- Green strength is increased by increasing the powder surface area.
- Green strength reduces with an increase in apparent density of the powders.

Properties of Sintered Compact

Dimensional Change during Sintering

Changes in dimensions of metal powder compact can occur during the production process.

The ability to control or predict the changes is an important factor in the fabrication process.

The metal powder used, as well as various processing parameters determine whether there will be shrinkage, growth or no-net change in dimensions in the finished metal powder compacts.

The size change, expressed as a % is defined as the linear difference between the die size and the size of the sintered part after cooling to room temperature.

Occasionally, the size change is also expressed as the difference between the green part size and the size of the sintered, cold P/M part

$$\text{rinkage (growth) \%} = \frac{\text{Change in length}}{\text{Sintered length}} \times$$

$$\text{rinkage (growth) \%} = \frac{\text{Change in length}}{\text{Unsintered length}}$$

A rectangular specimen (1¼" long and ¼" thick) or a hollow cylindrical specimen (1" outer diameter ¾" inner diameter & 1" long) is used for performing dimensional change test.

Average value of three test specimens is taken.

- **Sintered Density** - Method is similar to the one used for finding green density.
- **Porosity** refers to the open volume in a P/M part after sintering.
- It is very difficult to produce P/M part without porosity even after sintering.
- Sometimes porosity is deliberately engineered into the component to satisfy certain requirement as in filters and bearings.
- **Porosity can be classified into open and closed porosities.**
- In closed porosity, the individual pores are isolated within the material
- Open porosity - interconnected pores represent a network of pores that are connected with one another and to the external surface of the component.

- The total porosity present in the sintered part may be calculated from the following relation

$$\rho = 1 - \frac{\rho_p}{\rho_s}$$

where

ρ is the fractional porosity

ρ_p is the density of sintered product, and

ρ_s is the density of solid material